

Def.

$$y = e^x \Rightarrow y' = e^x.$$

2/3 32 3/4
3/4 1/4

$$y = a^x \Rightarrow y' = a^x \ln a.$$

P.f.

$$y = e^x$$

$$y' = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = \lim_{h \rightarrow 0} \left(e^x \cdot \frac{e^h - 1}{h} \right) = e^x.$$

$$y = a^x$$

$$y' = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} = \lim_{h \rightarrow 0} \left(a^x \cdot \frac{a^h - 1}{h} \right) = a^x \ln a.$$

Def.

$$y = \ln x \Rightarrow y' = \frac{1}{x}$$

$$y = \ln a^x \Rightarrow y' = \frac{1}{x \ln a}$$

P.f.

$$y = \ln x$$

$$y' = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \cdot \ln \frac{x+h}{x}$$

$$y = \ln a^x (= \frac{\ln x}{\ln a} = \frac{\ln x}{\ln a}) = \lim_{h \rightarrow 0} \ln \left(1 + \frac{h}{x} \right)^{\frac{1}{h}} = \frac{1}{x} \ln a$$

$$y' = \left(\frac{\ln x}{\ln a} \right)' = \frac{1}{\ln a} - (\ln x)' = \frac{1}{\ln a} - \frac{1}{x} = \frac{1}{x \ln a}$$